Classifying Financial News with Support Vector Machines

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Abstract

Our project aims to assist financial firms through the sentiment classification of financial news. We chose to use an SVM with TF-IDF vectorization to classify headlines as positive, negative, or neutral, due the clear separation boundary an SVM provides. The dataset we chose was FinancialPhraseBank, an open dataset with financial headlines labeled as positive, negative, or neutral from the perspective of a retail investor. We had to use balanced accuracy metrics, like the F1 score to evaluate our performance, since our dataset was unbalanced, and had much more neutral labels than positive or negative. We were able to achieve over 90% F1 accuracy through the SVM model and dataset, but also acknowledge the potential bias of the dataset towards the Finnish financial landscape. In the future, we plan to gather datasets from other countries to reduce bias, and also to have a more balanced label count in each category.

Introduction

People often make financial decisions influenced by what they see in news headlines. This means that in the short term, the market condition can be quite unpredictable and cannot be modeled by traditional economic models. For financial trading firms to keep their competitive edge on the market, it is crucial that they are able to factor in the general sentiment of the market when making a more informed financial decision. We introduce a machine learning model trained on a sentiment analysis dataset that is capable of classifying financial headlines, as positive, neutral or negative.

Method

To classify news headlines into positive, neutral, and negative sentiments, we use Support Vector Machines, with TF-IDF vectorization to remove unrelated financial terms that could be prevalent in all headlines. We chose support vector machines over other models like KNNs, because they have a very clear and well defined decision boundary. Additionally, we can use the kernel trick to great effect, since text data has high dimensionality [3]. After using GridSearchCV to optimize our model's hyperparameters, it found that the best kernel was a sigmoid kernel.

We also removed stop words in order to improve our model's accuracy. In all data, there are unimportant words or phrases that don't benefit us if we process them, and only serve to add noise. If we identify these words properly and remove them, we can raise the accuracy of our model. We identified stop words by getting familiar with our data and noticing some formatting patterns that would have increased the number of ineffective features. Our stop words include but are not limited to: “the”, “ .”, “ ‘s ”, and “, “. We tried using SMOTE as an oversampling method to handle our unbalanced dataset, but found that it surprisingly worsened performance. Thus we’ve elected to not use oversampling.

Data

Our dataset is FinancialPhraseBank [1], which contains sentiments for financial news headlines from the perspective of a retail investor. The headlines were labeled by 16 annotators, and the dataset has subsets based on how many annotators agreed with the majority opinion: the entire dataset is 50% agreement, and there are further subsets for 66%, 75%, and unanimous agreement [2]. It is an unbalanced dataset, where the majority of the data (59%) is neutral, 28% is positive, and only a small portion is labeled as negative sentiment (12%). Because of this unbalanced data, we can't look at the raw accuracy, and instead must use metrics like Balanced Accuracy, Recall, Precision, and F1 which is a combination of Recall and Precision. The use of these metrics allows for a more diverse output and a more balanced understanding of our model’s accuracy by balancing class distribution, or mitigating false positives and negatives. Our results included all four of these metrics.

Results

The model was measured using the performance metrics of balanced accuracy, precision, recall, and F1 score. These metrics were chosen to provide a comprehensive view of the classifier’s performance across different levels of agreement in the dataset.

The model was trained on the set with 50% agreement and tested across subsets with varying agreement levels, resulting in the following metrics:

Dataset Agree %	Balanced Acc.	Precision	Recall	F1 Score
50% Agree	0.6909	0.7555	0.7577	0.7557
66% Agree	0.8857	0.9256	0.9254	0.9246
75% Agree	0.8989	0.9335	0.9334	0.933
100% Agree (All Agree)	0.936	0.9519	0.9514	0.9516

The training dataset with 50% agreement acted as a superset, encompassing all other subsets. This likely made our model more efficient since it improved its ability to generalize across all subsets, but performance improved significantly when tested on datasets with higher agreement levels. In comparison, when the model was trained on the 100% agreement dataset, its balanced accuracy was only 80%.

Incorporating preprocessing techniques such as removing stop words improved our model's performance. For instance, precision and recall improved greatly in the higher agreement subsets, showcasing the benefit of reducing noise in textual data.

Since we had a sigmoid kernel, we couldn't just look at the coefficients to do analysis on how features were correlated with class. What we did instead was have the model evaluate a sentence consisting of each feature alone and report the probabilities it belonged to each class. Our vectorizer allowed both unigrams and bigrams. The most negative features included things like "down from," "decreased to," "down," "sales decreased," "fell by," etc. Neutral features were mostly nonsensical phrases like "eur111 while," "2009 euro18," "275 as," and anything involving the word "development." The most positive features included "to increase," "up from," "increased by," "rose to," etc. It's clear that anything involving upward movement is positive while anything involving downward movement is negative.

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Most negative features:
down from: 0.999994997844634
decreased to: 0.9999501900397018
down: 0.9999330130989482
decreased by: 0.9998878276137548
decreased: 0.9997350904513326
sales decreased: 0.9993482208975443
profit decreased: 0.9991701404179532
from decreased: 0.9991302492336496
fell by: 0.9989465906316962

Most neutral features:
development: 0.9990652825169073
eur111 while: 0.9988263049776943
eur1 275: 0.9981954961163552
2009 euro18: 0.9979428712840605
275 as: 0.9977610847738524
research development: 0.99749211260115
business development: 0.9972027687591478
euro18 million: 0.9971845721771491
development projects: 0.9964825131225189

Most positive features:
to increase: 0.9999998279894339
up from: 0.9999998149361452
increased by: 0.9999983220250511
increased: 0.9999982372510823
increase: 0.9999967734833586
an increase: 0.998010793363779
rose to: 0.9978698882505659
increase from: 0.9975728603257092
signed an: 0.9967674495007144
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We also created a script to test our model on some synthetic sentences and report the probabilities for each class. The results were somewhat disappointing: for example, the real headline "Dow slides for a seventh straight day for longest losing streak since 2020" from CNBC [4] has a 6% negative probability, 77% neutral, and 16% positive. This suggests that there could be some overfitting happening on our dataset.

Conclusion

In summary, we created an SVM model that can classify the sentiment of financial news headlines as positive, negative, or neutral with around 95% F1 accuracy, despite working with an unbalanced dataset. We believe this will be highly useful for financial trading firms that need to make decisions based on the current sentiment of the market. Since this dataset was made by Finnish researchers, there's a chance that the dataset is biased towards the state of the Finnish market. In the future, we could work on obtaining more data points for positive and negative labels to make a more balanced dataset, along with sampling data from a multitude of countries.

Contribution Chart

Everyone in the group did an excellent job and helped out with multiple subtasks. For the sake of brevity, only the main assignee for each subtask is mentioned.

Task	**Name and Student ID**	**Commentary**
Research Premise and Timeline for Report and Presentation	Alvin Yu (02039560)	Alvin did research on the issue with traditional economics models and suggested sentiment analysis. He also wrote the abstract, introduction, and method of both the presentation and the report. He also proposed a timeline for the project.
Data Collection, Presentation and Report Analysis, and Conclusion.	Shakib Tidjani (01957789)	Shakib collected datasets regarding financial sentiment analysis. He also wrote the conclusion and proposed follow up research that we could do in the future.
SVM Model Training	Tristan McDermott (02012319)	Tristan built an SVM model using the Python library scikit-learn and hosted the Github repository for the code. He also built a GridSearch to find the most optimal hyperparameters.
Data Analysis for Report and Presentation	Quang Tran (01991459)	Quang compiled data from our model into the report. He also helped with the interpretation of data through graphs.
Improving SVM Model Accuracy	Ephraim Acquah (02001407)	Ephraim added hyperparameters to improve the model and allowed our accuracy to surpass 90%. He also improved the accuracy

		further by using a higher quality labeled dataset and adding TF-IDF stop words.

References

[1] https://www.kaggle.com/datasets/ankurzing/sentiment-analysis-for-financial-news
[2] Malo, P., Sinha, A., Korhonen, P., Wallenius, J., & Takala, P. (2014). Good debt or bad debt: Detecting semantic orientations in economic texts. Journal of the Association for Information Science and Technology, 65(4), 782-796.
[3] Kowsari, K., Meimandi, K., Heidarysafa, M., Mendu, S., Barnes, L.E., & Brown, D.E. (2019). Text Classification Algorithms: A Survey. *ArXiv, abs/1904.08067*.
[4] https://www.cnbc.com/2024/12/12/stock-market-today-live-updates.html